Omar Elsherbiny

AEROSPACE ENGINEER · MUNICH, GERMANY

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PROFILE

Aerospace engineer in the Orbital Systems department at deltaVision, working on in-orbit refuelling and multi-orbit vehicles for dynamic space operations, including OPTRA, a propellant transfer demonstrator I helped design, analyse, build and test. My work sits between mechanical design and systems engineering: physical architecture, CAD and FEA, traced through to system-level requirements. MSc Aerospace (TUM) with a master's thesis on ENCORE, a self-refuelling ESA L-Class mission concept to Enceladus, and five conference papers at IAC and SSEA.

EXPERIENCE

deltaVision · Engineer, Orbital Systems — Munich, Germany · 04/2026 – present

- Working on multi-orbit vehicles for dynamic space operations: spacecraft that refuel in orbit so they can manoeuvre freely between orbital regimes instead of rationing propellant.
- Physical architecture, CAD design and structural analysis (FEA) for the Refuelling Module, the ESA GSTP-funded propellant transfer node that makes those operations possible.
- OPTRA, the Orbital Propellant Transfer Demonstrator (delivered, awaiting launch): redesigned the fluidic system configuration and the component clamping.
- Analysed the OPTRA structure with static FEA and built a dynamic model of the fluidic system and structure to simulate the random vibration environment.
- Assembled the demonstrator and ran its test campaign: pressure, leak, safety and functional tests, plus quasi-static load (sine burst), sine sweeps and random vibration.

Chair of Space Mobility and Propulsion, TUM · Researcher, master's thesis — Munich, Germany · 10/2025 – 03/2026

- Designed ENCORE, a candidate ESA L-Class mission to Enceladus with an orbiter and hopping lander, under Prof. Dr.-Ing. Chiara Manfletti.
- Built the trajectory (~7.5 km/s total Δv, Titan resonant capture saving ~650 m/s) and sized a hybrid chemical-electric water electrolysis propulsion system.
- Results presented at SSEA 2026 and in two papers at IAC 2026.

Rocket Factory Augsburg · Systems Engineering Intern — Augsburg, Germany · 06/2025 – 08/2025

- Built a risk assessment tool for safety-critical launcher systems.
- Ran an FMEA on the stage separation system and a COPV safety analysis.
- Reviewed 21 ground launch systems.

Professorship of Lunar and Planetary Exploration, TUM · Scientific Student Assistant — Munich, Germany · 05/2024 – 06/2025

- Built lab infrastructure and teaching demonstrators, including the PSR demonstrator, under Prof. Dr. Philipp Reiss.
- Semester thesis: designed and simulated (COMSOL) a cold plate and modular thermal shroud for a dirty thermal-vacuum chamber.

Coventry University · Researcher — Cairo, Egypt · 03/2022 – 05/2023

- Forecast spot and futures carbon emission prices with neural networks and multifractal detrended fluctuation analysis (MATLAB, Python), under Prof. Stratis Kanarachos.

Coventry University · Research Intern — Cairo, Egypt · 06/2021 – 06/2022

- Fabricated 3D-printed Wood PLA and ABS specimens in four designs and ran tensile tests to ISO standards.

Earlier internships — 2020 – 2022

- **Edibon International**, Madrid (09/2022): testing and validation of hydrogen fuel cell, wind turbine and pipe heat-transfer rigs.
- **Elsewedy Electric PSP**, Dubai (07–08/2021): operations at El Layyah power plant across engineering, procurement, construction, HSE and QC.
- **GASTEC**, Cairo (06–07/2021): CNG vehicle conversions, station maintenance and SolidWorks simulation.
- **Coventry University**, Cairo (10/2020 – 05/2021): manufacturing assistant (CNC, laser cutting, 3D printing) and mathematics tutor.

EDUCATION

MSc Aerospace, Technical University of Munich (TUM) — Munich, Germany · 10/2023 – 03/2026

- Final grade **1.5**, thesis **1.3**, both "very good" (German scale: 1.0 best, 4.0 pass).
- Master's thesis: Design and Analysis of an ESA L-Class Mission to Enceladus and the Saturnian System (ENCORE).
- Semester thesis: Design and Analysis of a Cold Plate and Thermal Shroud for a Dirty Thermal-Vacuum Chamber.
- Selected courses: Spacecraft Design, Space Mission Design, Systems Engineering, Model-Based Systems Engineering, Rocket Propulsion I & II, Electric Space Propulsion, Spacecraft Thermal Analysis and Test, Orbit and Flight Mechanics, Topology Optimization, Space Resources.

BEng Mechanical Engineering, Coventry University Coventry, UK · 09/2019 – 06/2023

- First Class Honours. Top Achiever Scholarship and Academic Scholarship.

PUBLICATIONS

1. F. Riegelsberger, **O. Elsherbiny** et al. (2026). ENCORE: Water Electrolysis Propulsion System Concept for a Self-Sustaining Enceladus Explorer Mission. *77th International Astronautical Congress*, Antalya. IAC-26,C4,9,4.
2. F. Riegelsberger, **O. Elsherbiny** et al. (2026). ENCORE: Concept for a Self-Sustaining Enceladus Mission. *77th International Astronautical Congress*, Antalya. IAC-26,A3,5,7.
3. F. Riegelsberger, **O. Elsherbiny**, C. Manfletti (2026). ENCORE: Mission Concept for Large-Scale Enceladus Exploration. *5th Symposium on Space Educational Activities*, TUM Ottobrunn. doi:10.14459/2026md1857625
4. **O. Elsherbiny**, M. Eltayeb, K. Zaki, O. Daoud, A. Abdellatif (2024). Development of an Air-Breathing Electrostatic Thruster for High-Altitude Propulsion. *75th International Astronautical Congress*, Milan.
5. A. Abdellatif, **O. Elsherbiny** (2024). Microgravity Testing of a Newly Developed Air-Breathing Electrostatic Thruster. *75th International Astronautical Congress*, Milan.

SELECTED PROJECTS AND ACTIVITIES

- **Water electrolysis thruster hot-fire tests** (TUM, 2026): nine hot fires of a gaseous H_2/O_2 thruster; combustion efficiency up to 85%; the test data showed where lumped-capacitance and Bartz models fall short for this chamber, motivating a transient thermal model.
- **Project Nectar** (TUM, 2024–25): reusable refueller for an Aldrin Earth-Mars cycler; did the CAD, physical architecture, structures, thermal and ConOps, plus requirements and MBSE; topology optimisation cut the frame mass by 24% (278 → 211 kg).
- **Upper-stage methalox engine** (TUM, 2025): preliminary design of a 1 MN gas-generator methalox engine and a two-stage launcher sized for 10 t to LEO.
- **Miniature InSight lander thermal test** (TUM, 2024–25): Thermal Desktop model tested in vacuum and correlated with a neural network.
- **Air-breathing electrostatic thruster** (Coventry, 2023): designed multi-stage thrusters and tested them on the ground and in a parabolic flight; two IAC 2024 papers.
- **Mars on Earth Biosphere** (TUM, 2023): first-place design of a test bed simulating Martian soil, pressure, temperature and atmosphere.
- **WARR Mission Control** (TUM, 2023–24): mission control team of TUM's student rocketry and spaceflight group.
- **Coventry robotics club and Minesweepers** (2021–23): technical officer of the robotics club; mechanical lead of the demining robot for the international Minesweepers competition.
- **Shell Imagine the Future** (2021): 3rd place in Egypt for a 2050 clean-energy city scenario.

SKILLS

Design and analysis	SolidWorks, Autodesk Inventor, Fusion 360, Abaqus, COMSOL, Thermal Desktop, Ansys CFX and Fluent, STAR-CCM+, Converge CFD
Test	Hot-fire testing; pressure, leak and functional testing; vibration testing (quasi-static/sine burst, sine sweeps, random)
Mission and systems	ASTOS, GMAT, NASA CEA, Astah SysML (MBSE), FMEA, requirements management
Programming	MATLAB, Simulink, Python, Julia
Manufacturing	CNC machining, laser cutting, 3D printing
Languages	Arabic (native), English (C1), German (A2), Spanish (A1)

REFERENCES

Prof. Dr. Philipp Reiss
Head of Lunar and Planetary Exploration, TUM

Fabian Riegelsberger
Research Associate, TUM Chair of Space Mobility and Propulsion

Kiran Chincholi
Head of Systems Engineering, Rocket Factory Augsburg

Akram Abdellatif
Senior Certification and Verification Engineer, Boeing